An Open-Source Quality Management System

1. Executive Summary

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2. The Problem

2.1 The Reproducibility Crisis

Academia is struggling with a reproducibility crisis. Published estimates of irreproducibility range from 51% (Hartshorne and Schachner 2012) to 89% (Begley and Ellis 2012) and it is estimated that irreproducible preclinical research wastes approximately 28 billion dollars a year in the US alone (Freedman, Cockburn, and Simcoe 2015).

Many of the systemic causes of irreproducibility can ultimately be traced to an underlying lack of a standards and best practices framework (Freedman, Cockburn, and Simcoe 2015; Global Biological Standards Institute 2013) or to poor data management (Soundararajan and Mishra 2023). Additionally, while academia has peer review for *results*, it rarely peer reviews *process*.

2.2 The Translation Gap

Furthermore, translating validated and reproducible results into the clinic is a daunting and difficult task, known as "the valley of death" (Coller and Califf 2009).

Academic institutions lack the experience and infrastructure needed to produce regulated products, or evidence supporting them, in a regulatory-compliant manner (Coller and Califf 2009; Stamler, Taber, and Califf 2003).

Data are assessed for fitness for use and reliability before they can inform a regulatory decision, and protocols and analysis plans are expected to be finalized before analyses are conducted (U.S. Food and Drug Administration 2023). Consequently, data originating in academia which have not been generated under prospectively defined and documented procedures cannot serve as the primary basis for a determination of substantial evidence (United States 2026). At best they can provide corroborative support, and only where they are "carefully conducted and documented". Reports lacking the details that permit scientific evaluation are not considered at all (United States 2026).

2.3 The Shared Root Cause

The reproducibility crisis and the translation gap are usually treated as separate problems, but they share a root cause. Reproducibility is a question of *internal repeatability*: can the work be repeated, by the same lab or another, and produce the same result? Translatability is a question of *external acceptability*: can someone outside the lab, such as a sponsor or a regulator, trust how the work was done well enough to rely on it? Both depend on research being conducted under defined, documented and controlled processes. They are the same requirement viewed from different directions.

A lack of consistent research quality systems is one of the systemic causes of irreproducibility, and there are few standards for quality management in life science research (Global Biological Standards Institute 2013). Standards are expected to make research outcomes "credible, reproducible, and translatable" alike (Global Biological Standards Institute 2013), and the FDA encourages sponsors to take a risk-based quality management approach to study oversight (U.S. Food and Drug Administration 2023).

Therefore one set of controls can serve both ends. Standard quality management processes such as document control, versioning, risk management, corrective and preventive action (CAPA), and change control make research repeatable inside the lab, and they produce the documented record that a sponsor or regulator needs to see from outside it.

2.4 The Cost Barrier

Even where academic institutions recognize the need for a quality system, the cost of establishing one is often prohibitive. Historically, academia has not had dedicated quality assurance personnel and has had difficulty funding these positions (Adamo et al. 2014). Academic laboratories face a "lack of funding for compliance costs" and high staff turnover due to a dependence on students as temporary personnel (Timóteo et al. 2021; Littrell et al. 2020). As a result, the cost of implementing quality management is often a deterrent (Littrell et al. 2020).

The published figures bear this out. The University of Kentucky estimates that building its GLP-compliant infrastructure cost $623,000 over three years, nearly half of which went to gap analyses, external consulting and training (Littrell et al. 2020).

Commercial electronic QMS (eQMS) software adds to these costs. Vendors rarely publish prices, but licenses typically run to tens of thousands of dollars a year even for a small team, and the software arrives as a largely empty, configurable platform. The SOPs, forms and workflows that make it a quality system must still be written, configured and validated, usually by consultants at additional cost.

Investment in quality is not wasted. At the University of Kentucky, industry-funded research more than doubled after the quality system was established, and annual maintenance fell to about 9% of the initial cost (Littrell et al. 2020). But the investment comes up front, and few academic labs can absorb it. Quality systems therefore remain largely confined to industry and a small number of well-funded academic facilities.

3. The Solution

We propose a free, lightweight, open-source quality management system for academic research, built on git, the version-control system researchers already use for code. Because the controls that make research reproducible are the same controls that make it submission-ready, a single QMS can address both the reproducibility crisis and the translation gap.

4. The Benefits of an Open-Source QMS

An open-source QMS removes the licensing cost, but being open source does more than that. It changes how the QMS is built, maintained, and trusted.

4.1 Eliminating Duplicated Effort

Today, an institution that needs a QMS largely builds its own. Each one commissions its own gap analysis, engages its own consultants, and writes its own SOPs, forms and workflows. At the University of Kentucky, nearly half of the initial cost of its GLP-compliant infrastructure went to gap analyses, external consulting and training (Littrell et al. 2020), and every other institution must pay for the same work. Yet the result is largely the same from one site to the next: the regulations being met are the same, and so are most of the controls needed to meet them. The cost of building a QMS is paid many times over for what is essentially the same artifact.

Commercial eQMS vendors do not remove this duplication. They monetize it. As described in Section 2.4, each customer pays for a largely empty platform and then pays again to have it filled with SOPs, forms and workflows much like those every other customer has already paid for.

Sharing what has already been built changes this. The University of Kentucky's Good Research Practice Resource Center, for example, provides templates for building a quality management system through consultation and training with the researchers it supports (Littrell et al. 2020). An open-source QMS extends that model beyond a single institution. The core is written once, maintained collectively, and adapted locally, so each new site starts from a working system rather than a blank page.

4.2 Best Practices Become Reviewable and Contestable

Today, the knowledge of how to interpret regulations and turn them into working controls is largely tacit. It lives in the heads of experienced quality staff and in the files of individual institutions, and it is most often obtained by hiring consultants. At the University of Kentucky, the GLP implementation relied on "specialized external consultants who have regulatory expertise" (Littrell et al. 2020). The resulting interpretations are rarely published, so they cannot be compared, cited, or challenged.

A public repository changes this. Every SOP, and every interpretation it embodies, is visible to anyone. Each version is recorded, so a site can cite exactly which version it adopted. Anyone can raise a concern or propose an improvement, and the discussion is recorded alongside the change it led to. Interpretations that are wrong or outdated can be challenged in the open rather than repeated in private.

This is peer review applied to *process*, which Section 2.1 noted academia rarely has. It is also how successful shared standards have been built before. The World Wide Web Consortium and the Internet Engineering Task Force steered competing organizations toward common standards through open, collaborative work, and the life sciences need a similar "dynamic, collaborative effort to standardize common scientific processes" (Freedman, Cockburn, and Simcoe 2015). An open-source QMS gives the field a place to do that work, and to converge over time on controls the community as a whole has reviewed.

4.3 Rationale Is Preserved, Not Just the Requirement

Boilerplate SOPs record what must be done. They rarely record why. The reasoning behind a control, such as the regulation it answers, the alternatives that were rejected, or the problem it was written to fix, usually lives only in the memory of the people who wrote it. In academia, where staff turnover is high (Section 2.4), those people are often gone within a few years, and the reasoning goes with them. The control then becomes something to follow without understanding, or to quietly drop.

A QMS built on git keeps that reasoning. Every change is recorded with who made it, when, and why, and the discussion that led to it stays linked to the change itself. The same applies to the QMS's own reviews. Gap analyses, the findings they raise, and the remediations that close them become part of the public record rather than a report filed away and forgotten.

This matters for audits as well as for continuity. Regulators expect changes to be explained, not just made. FDA guidance, for example, asks that revisions to a study protocol be "date-stamped," with "the rationale for each change" provided (U.S. Food and Drug Administration 2023). A QMS that carries its own reasoning can answer the question that both an auditor and a new staff member will eventually ask. Why do we do it this way?

4.4 Network Effects

Each site that adopts a common QMS makes it more valuable to every other site.

For sponsors and regulators, a shared framework means evaluating one QMS rather than many. Today, a sponsor considering an academic partner, or data generated in academia, must assess each institution's quality system on its own terms. If sites share a common, public QMS, the sponsor can review it once and then focus on how each site has adapted and followed it. Familiarity itself has value. The University of Kentucky reports that having "a QM framework familiar to the pharmaceutical/device industry" broadened its industry collaborations (Littrell et al. 2020). A QMS shared across many institutions extends that familiarity to all of them, which directly narrows the translation gap described in Section 2.2.

For multi-site studies, a shared QMS means sites start from the same procedures. Irreproducibility fundamentally "stems from undefined variance in reagents, practices, and assays between laboratories" (Global Biological Standards Institute 2013). Sites that follow the same documented procedures remove one source of that variance by default, rather than negotiating common procedures for each new collaboration.

These benefits compound. Every site that adopts the QMS adds to the pool of reviewers, contributors and evidence that it works, and makes the QMS more attractive to the next site. With a commercial eQMS,

each new customer adds revenue for the vendor but nothing for other customers. With an open-source QMS, every new adopter increases its value to all the others.

4.5 A Teaching Artifact

Academic investigators often "lack experience in drug development and regulatory processes" (Coller and Califf 2009), and there are few accessible ways to gain it. Regulations and standards are written for compliance, not instruction, and the working QMSs that show how they are applied in practice are held privately by the companies and institutions that built them.

An open-source QMS is readable by anyone. A student or new investigator can see how an abstract requirement becomes a concrete procedure, form and record, and, through the history described in Section 4.3, why it was written that way. Because it can be copied and modified freely, it can also be practiced on. A lab can trial a procedure, or a course can build exercises around it, without any risk to a live quality system.

Academia is, above all, a training environment. The high staff turnover that makes quality systems hard to sustain is "inherent to the training environment of academia" (Littrell et al. 2020), which also means a steady stream of students and trainees passes through every lab. Institutions that conduct regulated research already value the opportunity "to teach students, staff, and faculty how to perform regulated research" (Adamo et al. 2014). An open-source QMS lets every institution offer that, whether or not it runs regulated studies, and sends trainees on to their next position already literate in quality management.

4.6 Extensibility Across Domains

Most of a QMS is not specific to any one kind of research. Document control, change control, training records, CAPA and internal audit work the same way whether a lab is running an observational study or developing software. The QMS is therefore structured as a domain-agnostic core, with domain-specific overlays that add the procedures, forms and regulatory mappings a particular field needs. A new domain does not need a new QMS, only a new overlay.

The two domains already built, software medical devices and observational studies (Section 5), are quite different from each other, which suggests the core generalizes. Candidate future domains include imaging core facilities, biobanks, AI/ML model development and wet-lab protocols, each of which faces reproducibility and quality pressures of its own.

Medical device startups are a particularly natural extension. Many begin as academic spinouts, carrying a discovery out of the lab and into the valley of death described in Section 2.2. Discoveries made at academic medical centers are typically handed to industry for development, yet "early stage (biotechnology) entities often are poorly suited to this task" (Stamler, Taber, and Califf 2003). A device company generally needs a QMS aligned with ISO 13485 to bring a product to market, and it needs one at the point when it can least afford one. The existing software medical device QMS (Section 5) already covers much of this ground, and a device overlay built on the same core would let a spinout carry its parent lab's quality system with it rather than starting again from scratch.

5. Prior Work

6. Implementation: QMS as Code

6.1 Minimal Primitives

6.2 Deliberate Vendor Neutrality

6.3 Cost

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